

Literature Review: ARIA - Activity Recognition & Inference Architecture

A Multi-Human Activity Detection & Role-Based Tracking System
in a Shared Service Counter Environment

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1. Introduction

ARIA proposes a privacy-preserving, multimodal system for detecting and attributing the activities of multiple staff members operating simultaneously in a shared service counter environment. It combines laptop-based computer vision (YOLOv8 + ByteTrack + MediaPipe), ambient audio analysis, and supplementary IoT sensing (Arduino ESP8266) to produce role-labelled activity data in real time, with inference running entirely at the edge and only statistical feature vectors forwarded to the cloud.

This literature review evaluates six bodies of work that collectively ground ARIA's design: studies validating multimodal sensing and edge inference, a collaborative edge-cloud architecture for multi-user recognition, a healthcare survey justifying the audio-vision fusion strategy, a dataset construction methodology, a multi-object tracking algorithm, and a survey identifying the multi-subject attribution gap that ARIA targets. Rather than summarising these works in isolation, the review draws cross-cutting connections, highlights convergences, identifies conflicting constraints, and maps each source to specific design decisions within ARIA.

2. Multimodal Sensing and Edge Inference

2.1. Nakabayashi & Saito (2024) – Multimodal HAR on Edge Devices

Nakabayashi and Saito demonstrated that fusing camera images with IMU data (acceleration and angular velocity) within a lightweight attention-mechanism network achieves approximately 97% precision for classifying five activity classes on the UESTC-MMEA-CL dataset, subsequently outperforming single-sensor baselines. Critically, the network was designed specifically for edge devices with limited computational resources, proving that high-accuracy multimodal inference does not require cloud offloading.

This work is directly foundational to ARIA's architecture. It validates two of ARIA's core bets: first, that fusing heterogeneous sensor streams (vision, audio, IoT) outperforms any single modality, and secondly, that a compact, efficient model can perform this fusion locally without server-side compute. The attention mechanism that dynamically weights sensor contributions depending on input quality is also conceptually aligned with ARIA's feature-vector fusion, where PIR, ultrasonic, audio amplitude, and skeleton keypoint features are combined before classification.

However, the single-user scope of Nakabayashi and Saito represents its most significant limitation relative to ARIA's requirements. The paper does not address identity disambiguation when two or more people share the frame, nor does it consider role-differentiated attribution. This gap – that high-performing single-user multimodal HAR breaks down under multi-occupancy – is precisely the extension that ARIA is designed to provide.

2.2. Golipoor & Sigg (2024) – RFID-Based HAR Using Multimodal CNNs

Golipoor and Sigg introduced a model for human activity recognition that leverages battery-less, body-worn reflective antenna components combined with multimodal convolutional neural networks, targeting industrial and factory automation settings where safety, efficiency, and productivity monitoring are important. Their work demonstrated that non-visual passive sensing can be combined with neural classification pipelines to recognise activities in professional environments, a setting closer to ARIA's service counter domain than the healthcare or domestic contexts that dominate most HAR literature.

This paper is relevant to ARIA on two levels. First, it validates the use of passive, ambient sensors, rather than cameras alone, as a viable strategy for workplace activity recognition, providing precedent for ARIA's hybrid sensing approach combining PIR, ultrasonic, and audio modules. Second, it establishes that CNN-based classifiers are effective at processing heterogeneous, non-visual sensor inputs, supporting ARIA's choice of a 1D-CNN as one of its candidate classifiers.

The key contrast with ARIA lies in sensing modality: RFID requires wearable tags attached to participants, introducing a cooperation dependency. ARIA's design choice of entirely passive ambient sensing addresses this directly, making it deployable in naturalistic environments where workers are not expected to carry or wear tracking hardware.

3. Multi-User Activity Recognition Architectures

3.1. Li, Gravina, Savaglio & Fortino (2023) – BSN-Enabled Collaborative Edge-Cloud Architecture

Li et al. proposed a hybrid architecture in which body sensor networks (BSNs) perform local inference at the edge while model training is offloaded to the cloud, enabling multi-user activity recognition across distributed IoT participants. The paper demonstrated that separating the training phase (resource intensive and cloud hosted) from the inference phase (lightweight and edge-hosted) is both feasible and effective, achieving multi-user recognition without requiring all computation to occur at a central server.

This directly validates ARIA's three-tier architecture. ARIA's Tier 2 (laptop edge inference) mirrors the BSN inference layer, while ARIA's Tier 3 (Google Colab retraining + Firebase sync) mirrors the cloud training layer. The edge-cloud split is not merely a technical convenience in ARIA, it is a privacy-by-design decision. By transmitting only statistical feature vectors to the cloud rather than raw video, ARIA follows the same principle Li et al. established: meaningful inference can be performed locally, with only abstracted representations sent upstream.

The critical limitation of Li et al. is the wearable BSN dependency. Body sensor networks require participants to actively wear sensor hardware throughout the monitoring period, which is impractical for a passive service counter environment where staff members should not be required to carry monitoring devices. ARIA substitutes wearable sensors with ambient sensing (PIR, ultrasonic, microphone, webcam), making the architecture passive and non-intrusive. The edge-cloud paradigm is preserved; the sensing layer is reimaged.

3.2. Arrotta, Civitarese, Chen, Cumin & Bettini (2025) – Multi-Subject HAR Survey

Arrotta et al. published a comprehensive survey of multi-subject human activity recognition, formalising the problem and cataloguing its open challenges. The survey identifies that while single-subject HAR has achieved mature performance across many sensing modalities, multi-subject HAR remains an open and complex problem, particularly in scenarios where multiple occupants are engaged in simultaneous, potentially overlapping activities. The authors note a persistent shortage of multi-subject datasets, underdeveloped evaluation protocols for per-subject attribution, and the particular difficulty of activity disambiguation in shared spaces.

This survey performs a dual function for ARIA. On one hand, it provides authoritative confirmation that ARIA's target problem, role-differentiated activity attribution in a shared space, is indeed an acknowledged unsolved extension of the existing HAR literature, not a problem already addressed elsewhere. On the other

hand, it prescribes evaluation standards: per-class F1 scoring, confusion matrices, and leave-one-session-out cross-validation are the expected evaluation protocols for any credible multi-subject HAR system.

Arrotta et al. also underscore the value of new datasets. The survey highlights the scarcity of multi-subject annotated datasets as a major bottleneck for progress in the field. ARIA's secondary contribution, a labelled, publishable dataset of counter-staff activities in a shared space, directly responds to this identified gap.

4. Multimodal Fusion Strategy: Justification for Audio & Vision

4.1. Cristina, Despotovic, Pérez-Rodríguez & Aleksic (2024) – Audio- and Video-Based HAR Systems in Healthcare

Cristina et al. conducted a scoping review of audio- and video-based activity recognition systems in healthcare, covering complexity, performance, robustness, scalability, and privacy considerations. A central finding is that combining audio and video modalities significantly improves HAR accuracy over single-modality systems, particularly for activities with similar visual signatures. For instance, distinguishing seated-desk-work from seated-phone-conversations, where the visual difference is subtle but the audio signature differs substantially.

For ARIA, this paper provides systematic evidence for the audio-vision multimodal design. The activity taxonomy ARIA targets includes precisely such visually ambiguous pairs: a counter agent typing versus a counter agent speaking on the phone, or a station manager performing desk work versus a station manager in active conversation with a colleague. The audio channel, captured via a MAX9814 microphone module, provides discriminative signal for these cases that the skeletal keypoint stream alone cannot reliably separate.

However, the healthcare framing of Cristina et al. introduces an important constraint: their reviewed systems operate in single-patient or clearly separated multi-patient clinical environments, not shared occupancy spaces. Privacy and scalability findings from healthcare may not transfer directly to a workplace counter context. ARIA's architecture reflects this distinction by adopting ROI masking and operating entirely at the edge, measures not required in most clinical deployments but essential when cameras face towards staff in a semi-public workspace.

5. Dataset Construction Methodology

5.1. Moreaux, Ortiz, Ferrané & Lerasle (2019) – Kitchen20 Dataset for Audio-Based Action Recognition

Moreaux et al. introduced the Kitchen20 dataset, a raw-audio, environmental, kitchen-related, non-vocal dataset designed to fill a gap in domain-specific audio activity data. The paper contributed not only the dataset itself, but a replicable methodology: structured activity scripting, per-class audio segmentation, human annotation validation, and benchmark comparisons with established datasets such as ESC-50. The authors also proposed merging Kitchen20 with ESC-50 to form a larger ESC-70 benchmark, demonstrating how domain-specific datasets can complement general-purpose ones.

Kitchen20 is the most methodologically transferable work to ARIA's dataset contribution. ARIA's planned dataset, labelled multimodal sensor recordings of counter-staff activities, collected in a controlled simulated

environment, will follow the same core protocol: scripted activity sequences, per-class labelling, and documentation suitable for public repository submission. The specific parallels include activity scripting to ensure class coverage, timestamped annotation to align multimodal sensor streams, and data card documentation describing collection conditions, participant demographics, and sensor configurations.

The key difference is modality depth: Kitchen20 is audio-only and single-person, while ARIA’s dataset will be multimodal (vision keypoints, audio amplitude, PIR/ultrasonic readings) and multi-person. Kitchen20 illustrates what a well-constructed domain-specific audio dataset looks like. ARIA expands this template to a richer sensor stack and a more complex multi-occupant labelling problem.

6. Core Algorithmic Components

6.1. ByteTrack – Multi-Object Tracking by Associating Every Detection Box (Zhang et al., 2022)

ByteTrack, published at ECCV 2022, addresses a fundamental limitation in multi-object tracking: prior methods discard low-confidence detection boxes, which frequently correspond to occluded or partially visible people, generating fragmented trajectories. ByteTrack’s contribution is to associate every detection box, including low-score ones, with existing tracklets using motion similarity, achieving 80.3 MOTA, 77.3 IDF1, and 63.1 HOTA on the MOT17 benchmark at 30 frames per second on a single V100 GPU.

ByteTrack is ARIA’s chosen multi-person tracking backbone, integrated via the Ultralytics YOLOv8 tracking API. Its relevance is direct: in the service counter environment, a staff member may be partially occluded by a counter fixture, another person, or the camera angle. Without ByteTrack’s low-score association, such a person would produce a fragmented track, breaking the identity continuity required for per-individual activity attribution. ByteTrack’s real-time performance at 30 FPS also ensures that the tracking layer does not become a bottleneck in ARIA’s pipeline.

ByteTrack does not, on its own, perform activity classification; it provides stable, consistent identities across frames. The skeleton keypoint extraction (MediaPipe / YOLOv8-pose) and the downstream classifier then operates on these stable track IDs to assign activity labels. This layered separation of tracking from classification is a deliberate architectural choice that reflects ByteTrack’s design philosophy: reliable tracking as an enabler, not a classifier.

7. Synthesis and Critical Comparison

The six works reviewed span four functional dimensions of ARIA:

- 1. multimodal sensing and edge inference
- 2. multi-user architecture
- 3. fusion strategy justification, and
- 4. algorithmic components and dataset methodology.

The table below maps each source to its contribution and the gap it highlights.

Source	Domain	Core Contribution to ARIA	Gap ARIA Addresses
Nakabayashi & Saito (2024)	HAR / Edge AI	Validates multimodal fusion + edge inference; ~97% precision	Single-user only; no attribution in shared space

Li et al. (2023)	IoT / Edge-Cloud	Validates edge-cloud split for multi-user HAR	Wearable sensor dependency; not passive
Cristina et al. (2024)	Healthcare HAR	Audio + video fusion outperforms single modality	Healthcare framing; no multi-person attribution
Moreaux et al. (2019)	Audio datasets	Domain-specific dataset construction methodology	Audio-only; single-person; no multimodal fusion
Arrotta et al. (2025)	Multi-subject HAR	Identifies multi-subject attribution as an open problem	Dataset scarcity; no workplace domain coverage
Zhang et al. / ByteTrack (2022)	Computer Vision	Robust real-time multi-person identity tracking	No activity classification; tracking only
Golipoor & Sigg (2024)	Industrial HAR	Passive sensor + CNN for workplace activity recognition	Wearable RFID tags required; not ambient

A theme that cuts all seven works is the single-to-multi-person transition as an unsolved scaling problem. Nakabayashi and Saito achieve strong results in isolation but do not attempt multi-occupancy. Cristina et al.’s reviewed healthcare systems largely operate on one patient at a time. Moreaux et al.’s Kitchen20 is explicitly single-person. Even Li et al., who explicitly target multi-user recognition, rely on wearable BSNs that uniquely identify each participant, not solving the harder problem of attributing activities to passively observed individuals distinguished only by spatial position and visual appearance. Arrotta et al.’s survey makes this gap explicit: passive, camera-and-microphone-based multi-subject attribution in shared spaces is identified as a frontier challenge.

A second cross-cutting theme is the privacy-edge tension. Both Li et al. and Cristina et al. acknowledge that cloud video processing raises data protection concerns, with Cristina et al. noting scalability and privacy as two of the most pressing challenges in deployed HAR systems. The recent industry literature on GDPR-compliant edge AI reinforces this, demonstrating that edge-first architectures are not merely a performance optimisation but a regulatory necessity when processing individuals’ visual and audio data in semi-public spaces. ARIA’s design, with local inference, feature-vector-only cloud transmission, and ROI masking, represents a concrete implementation of the privacy-by-design principle that the reviewed literature collectively advocates but does not consistently realise.

A third theme, visible in comparing Nakabayashi and Saito with Golipoor and Sigg, is the trade-off between sensing modality intrusiveness and deployment practicality. Wearable systems (BSNs, RFID tags) can achieve high per-user accuracy because the sensor is physically attached to the individual, removing the ambiguity of who is being measured. Ambient systems must infer identity from appearance, position, or temporal patterns, a harder problem. ARIA’s choice of ambient sensing accepts this harder sub-problem in exchange for deployability in environments where staff cannot be required to carry hardware. ByteTrack’s contribution, maintaining persistent track IDs across occlusion events, is precisely what bridges this gap by providing stable visual identities without wearables.

Finally, Moreaux et al. and Arrotta et al. together frame ARIA’s secondary dataset contribution as independently significant. Moreaux et al. demonstrate that domain-specific datasets become reference benchmarks and enable downstream research communities to develop and compare approaches on a shared evaluation baseline. Arrotta et al. note that the scarcity of multi-subject datasets is itself a major bottleneck for the field. A labelled, publicly released, multimodal counter-staff activity dataset with full data card

documentation would be the first of its kind and would address both the Kitchen20-style gap (single-person, audio-only) and the Arrotta et al.-identified gap (multi-subject, workplace domain).

8. Conclusion

Across the reviewed literature, no existing system simultaneously achieves role-aware activity attribution, passive ambient sensing, multimodal sensor fusion, edge-first inference, and domain relevance to a service counter environment. Each source validates one or more of ARIA's design decisions while leaving the full combination unaddressed. The convergence of these gaps, confirmed by both primary research papers and a comprehensive survey, establishes ARIA as a necessary and well-motivated contribution to the multi-subject activity recognition field.

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